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ORIGINAL INVESTIGATION

Football Movement Profile–Based Creatine-Kinase Prediction Performs Similarly to Global Positioning System–Derived Machine Learning Models in National-Team Soccer Players

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Purpose: The relationship between external load and creatine-kinase (CK) response at the team/position or individual level using Global Positioning Systems (GPS) has been studied. This study aimed to compare GPS-derived and Football Movement Profile (FMP) –derived CK-prediction models for national-team soccer players. The second aim was to compare the performance of general and individualized CK prediction models. ***Methods:*** Four hundred forty-four national-team soccer players (under 15 [U15] to senior) were monitored during training sessions and matches using GPS. CK was measured every morning from whole blood. The players had 19.3 (18.1) individual GPS-CK pairs, resulting in a total of 8570 data points. Machine learning models were built using (1) GPS-derived or (2) FMP-based parameters or (3) the combination of the 2 to predict the following days' CK value. The performance of general and individual-specific prediction models was compared. The performance of the models was described by R^2 and the root-mean-square error (RMSE, in units per liter for CK values). ***Results:*** The FMP model ($R^2 = .60$, RMSE = 144.6 U/L) performed similarly to the GPS-based model ($R^2 = .62$, RMSE = 141.2 U/L) and the combination of the 2 ($R^2 = .62$, RMSE = 140.3 U/L). The prediction power of the general model was better on average ($R^2 = .57$ vs $R^2 = .37$) and for 73% of the players than the individualized model. ***Conclusions:*** The results suggest that FMP-based CK-prediction models perform similarly to those based on GPS-derived metrics. General machine learning models' prediction power was higher than those of the individual-specific models. These findings can be used to monitor postmatch recovery strategies and to optimize weekly training periodization.

Keywords: muscle damage, intertial sensor, training load, match load

Increasing physical demands of elite soccer have been suggested in previous studies examining different championships and time periods.[1–3] To monitor player readiness to perform, help with weekly periodization, and minimize injury risk, fatigue monitoring is common practice in elite soccer. One of the most widely used biomarkers to quantify muscle damage is creatine kinase (CK).[4] Elevated CK pretraining has been shown to impair training output including high-speed, very high-speed, and sprint distance, accelerations, decelerations, explosive distance, and maximal velocity.[5] It reaches its peak 14 to 48 hours after the match, and it takes 42 to 120 hours to return to prematch values depending on the activity level during the recovery process.[6]

The use of Global Positioning Systems (GPS) to monitor training and match load in elite soccer is commonplace.[7] In addition to a GPS receiver, most GPS units also contain inertial sensors (accelerometer, gyroscope, and magnetometer) to calculate additional parameters (eg, jumps, changes of directions, and PlayerLoad). Several studies have examined the relationship between GPS/inertial sensor-derived parameters and CK changes after soccer matches,[8–13] training sessions,[14] or the combination of both.[15–17] Most of these studies utilized small sample size (< 300 data points) and correlations between external load metrics and CK change.[8–13] Recent studies utilized more advanced statistical procedures[14–16] to predict CK from external load. Another practical question is the applicability of these findings in the real world. Even though some position and activity type-specific machine learning models performed relatively well ($R^2 = .50$–$.88$), some of the key training/match load metrics identified by the models were not practically relevant (eg, distance 0–0.7 km/h) or are currently not available for live monitoring (eg, inertial movement analysis [IMA] number of low-intensity left changes of directions, and IMA number of medium decelerations).[15] In the latter case, these parameters cannot be controlled and modified during the training session/match if necessary.

In addition, GPS signal quality might be compromised in big stadiums and is not available indoors. In these situations, GPS-derived metrics are not available, or data quality might be questionable. The Football Movement Profile (FMP) algorithm, developed by Catapult Sports, attempts to overcome the aforementioned limitations. FMP assigns all movements into 6 categories based on the intensity (very low, low, medium, and high) and the movement type (linear running or dynamic-change in direction or speed).[18] Consequently, the following metrics are available: very low duration (~standing), low duration (~walking), running medium duration (~steady-state linear and curvilinear jogging and running),

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running high duration (~steady-state linear and curvilinear running at high intensity), dynamic medium (medium-intensity multidirectional movements, and linear acceleration and deceleration), and dynamic high (high-intensity multidirectional movements, and linear acceleration and deceleration). In a cohort of elite youth soccer players, FMP highlighted that 27.8% of the match duration was spent in dynamic activities and the algorithm clearly differentiated in the movement pattern of different positions.[18] More recently large positive correlations were found between FMP parameters and CK change ($r = .62$ [.12]) in elite soccer players, and they could be clustered based on their CK sensitivity to FMP values.[17]

Most previous studies have examined general relationships between training/match load and CK change in soccer players.[8–13] Even though position-specific[15] and individualized models[16] seem to be more accurate, the question remains whether general or individualized models are more appropriate.

Therefore, the aim of this study was 2-fold. First, we aimed to examine whether the FMP-derived parameters can be as accurate as GPS + IMA-derived metrics to predict the following days' CK of elite soccer players using machine learning models. Second, we compared the prediction power and accuracy of general and individual-specific CK prediction models in this cohort.

Methods

Subjects

Four hundred forty-four elite male national team soccer players (from under 15 [U15] to senior level) were monitored between 2019 and 2022 in national-team training camps consisting of international friendly and competitive matches. Subject characteristics are as follows (mean [SD]): age 19.2 [4.7] years, height 181.7 [6.4] cm, mass 73.9 [7.7] kg. All players performed 5 to 7 training sessions and 1 to 2 matches per week. The players and their parents provided informed consent to participate in the study, which was approved by the ethical review board of the local university. The investigation conformed to the Code of Ethics of the World Medical Association.

Design

This study utilized an observational research design and was conducted between 2019 and 2022. Players took part in 4 different types of training camps (Figure 1). We linked the athletes' CK concentration with their external load on the preceding day of the CK measurement.[19] Data points (pairs) were omitted where either the CK concentrations or the external load parameters were not available. The aim was to build general and personalized machine learning models for each athlete which predicted the CK value of the athletes on the next morning, knowing the actual CK value and the training or match load on any given day.

External Load Monitoring

The activity profile of the players was monitored during each training session and matches with 10-Hz GPS units (Catapult S7 and G7 for field players and goalkeepers, Catapult Sports). In order to avoid interunit differences, each player wore the same unit during the study period. The GPS unit was worn in a custom-made vest between the shoulder blades and did not limit the movements of the upper limbs or the torso. The mean velocity difference of the S7 units was validated against the gold standard Vicon system for several velocity zones during multiple football-specific drills.[20] The GPS, IMA, and FMP parameters used in this study are detailed

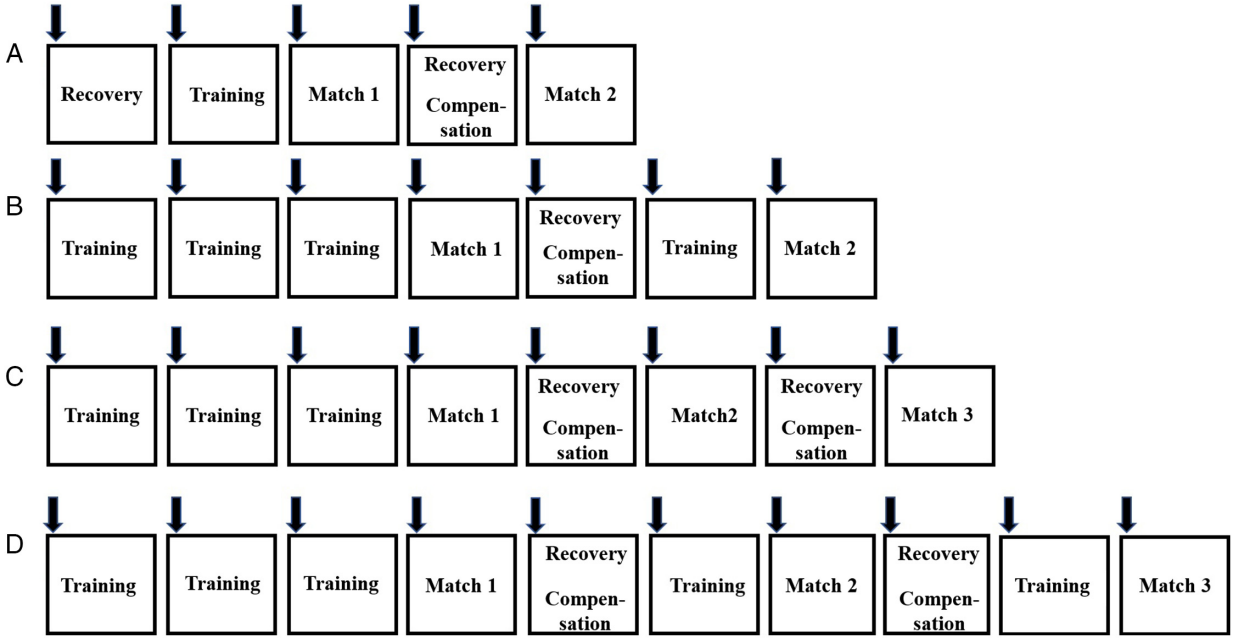

Figure 1 — Study design. The players followed 1 of these schedules in training camp. (A) After playing in the national championship in their clubs, they came to the national team on Sunday and had a recovery session. On Monday, they had 1 tactical training session followed by an international friendly game on Tuesday. The players who played >45 minutes completed a regeneration session on Wednesday, and the rest of the team had a compensation session. The training camp was finished with a second international friendly game on Thursday. (B) Typical schedule for the under-21 and national team. After some training sessions, the team played 2 international matches with 2 days in between. (C) During some training camps, the team played 3 international friendly matches with 1 recovery day in between. (D) During the European Qualification, European Championship, and World Cup, 3 to 5 international competitive matches were played with 2 recovery days in between. The black arrows indicate creatine-kinase measurements.

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in Table 1. The GPS units also provide IMA data based on the integration of 100-Hz microsensor data (accelerometer, gyroscope, and magnetometer). The reliability of this analysis and the description of the calculations have been published previously.[21]

Creatine Kinase

Capillary blood samples for CK determination were taken from fingertips after overnight fasting, every morning at a standardized time (8–10 AM) as described previously.[15,16] Whole blood was analyzed using a Reflotron Plus Clinical Chemistry Analyzer (Roche) between 2019 and 2021. The validity of this method was determined in a previous study.[22]

From 2022, Tascom SimplexTas 101 Immuno Chemistry System (Tascom) was used to determine CK from capillary blood samples. We assessed the validity of the SimplexTas machine to determine CK in 36 elite soccer players with 48 individual data points, which were compared against venous samples and analyzed in an accredited medical laboratory (Beckman Coulter AU5800 chemical analyses). A strong, positive correlation was found (Spearman $r = .954$; $P < .001$). The difference between capillary and venous samples was significant (Wilcoxon signed-rank test, $T = 382$, $z = -2.113$, $P = .035$, $r = .30$), and the median difference between the 2 machines was 14 U/L. The output of Reflotron and SimplexTas was compared using the samples of 45 youth and senior soccer players, resulting in 116 capillary blood sample pairs.[23] The CK results of the 2 machines correlated significantly ($r = .92$, $P < .05$), with significant difference between the results ($P < .05$; $t = 13.4$). Thereafter, a correction equation was created to convert the CK values measured by Reflotron Plus to SimplexTas 101.

Table 1 Parameter Groups, Parameters, and Parameter Definitions Used to Predict Creatine-Kinase of the Following Day

Parameter group	Parameter	Definition
GPS	Total distance	Distance covered in all velocity zones, m
	High-intensity distance	Distance above 19.8 km/h, m
	Sprint distance	Distance above 25.1 km/h, m
	Distance >30 km/h	Distance above 30 km/h, m
	Maximum velocity	Maximum velocity during a training session or match, km/h
	High metabolic power distance	Distance above 25.5 W/kg, m
	Number of accelerations	Number of accelerations between 2 and 4 m/s^2
	Number of decelerations	Number of decelerations between −2 and 4 m/s^2
IMA	Number of IMA events	Number of accelerations, decelerations, left and right changes of directions above 1.5 m/s
	Number of high-intensity IMA events	Number of accelerations, decelerations, left and right changes of directions above 3.5 m/s
	Number of high-intensity IMA accelerations	Number of IMA-based accelerations above 3.5 m/s
	Number of high-intensity IMA decelerations	Number of IMA-based decelerations above 3.5 m/s
	Number of high-intensity IMA left changes of directions	Number of IMA-based left changes of directions above 3.5 m/s
	Number of high-intensity IMA right changes of directions	Number of IMA-based right changes of directions above 3.5 m/s
	PlayerLoad	Total player load, AU
FMP	Very low duration	Standing-like movements, % of total time, duration, min
	Low duration	Walking-like movements, % of total time, duration, min
	Running medium duration	Steady-state linear and curved jogging and running at medium intensity, % of total time, duration, min
	Running high duration	Steady-state linear and curved running at high intensity, % of total time, duration, min
	Dynamic medium duration	Medium-intensity multidirectional movements, linear acceleration and deceleration movements; % of total time, duration, min
	Dynamic high duration	High-intensity multidirectional movements, linear acceleration and deceleration movements; % of total time, duration, min
	Total running duration	Running medium + running high duration; % of total time, duration, min
	Total dynamic duration	Dynamic medium + dynamic high duration; % of total time, duration, min
Individual average	Eg, total distance per player mean	The average total distance of all activities of a player until a certain activity, it was calculated for all parameters
Individual difference	Eg, total distance per player difference	Difference between the total distance of the actual activity and the average total distance of all activities until the actual activity for a certain player, it was calculated for all parameters

Abbreviations: FMP, Football Movement Profile; GPS, Global Positioning System; IMA, Inertial Movement Analysis.

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Statistical Analysis

Machine learning models were built to determine the best predicted output. Three predicted parameters were tested based on the actual days' CK value and training/match load: absolute CK value of the following day, absolute CK change from the actual day to the following day ($CK_{following\ day} - CK_{actual\ day}$), and CK % change: $[(CK_{following\ day} - CK_{actual\ day})/CK_{actual\ day}] \times 100$.

The prediction power of 3 types of load inputs was tested: GPS + IMA-derived parameters, FMP parameters, and the combination of both. The model was given some additional input variables to increase its accuracy: load type (training/match), name of the player to differentiate between individuals, position (goalkeeper, central defender, fullback, central midfielder, winger, and attacker), and age group. The feature importance was characterized by the shapley additive explanations (SHAP) value, which quantifies the contribution of each feature to a particular prediction outcome.[24]

For the players with >20 load-CK data point pairs, individual machine learning models were built. In this case, both FMP and GPS + IMA-derived parameters were used for the load input. To find the most accurate model with the best parameter set, walk forward cross-validation was used. The data set was sorted by time and split into 6 pieces. The training set consisted of the first 2 pieces, and the test set was the third piece. Parameter selection and model fitting were performed on the training set, and the model was evaluated on the test set. Thereafter, the test set was appended to the training set and the next piece of the data set was marked as the test set. This process was repeated until the entire data set was tested.

Both linear (Linear, Ridge, and Lasso Regression[25]) and complex machine learning models (Gradient Boosting Machine[26] and Random Forest[27]) were built. The predictive power of the models was characterized by the R^2 value (coefficient of determination) to assess the goodness of fit of a regression model.[28] The maximum is 1, which indicates a perfect model, a baseline model, which always predicts the average scores 0, and models that predict worse than the baseline achieve negative R^2. The mean absolute error (MAE, in units per liter), the root mean square error (RMSE, in units per liter), and the mean absolute percentage error (MAPE, in percentages) in predicted versus measured CK values described the accuracy of the prediction. All statistical analyses were performed in Python (version 3.10.6) using Sklearn (version 1.1.2) and Statsmodels (version 0.13.2). Results are reported as mean and SD.

Results

CK Data-Set Characteristics

The total number of individual player observations (where both CK and external load data were available) was 8570. The average number of data points per player was 19.3 (18.1).

Predicted Parameters

The initial models produced the best prediction for the absolute CK value of the following day ($R^2 = .574$, MAE = 110.8, RMSE = 162.3, MAPE = 0.49). The prediction of the absolute ($R^2 = .384$, MAE = 101.8, RMSE = 154.0, MAPE = 2.53) and relative CK changes ($R^2 = .335$, MAE = 59.2, RMSE = 195.6, MAPE = 2.25) produced lower prediction power and accuracy.

CK Prediction Model Using the Whole Sample

Table 2 shows the accuracy and the prediction power of the models using GPS + IMA-derived parameters, FMP parameters, or the combination of both. Figure 2A, 2B, and 2C shows the predicted versus actual CK values in the test data set (2571 data points out of 8570) based on GPS + IMA-derived, FMP parameters, or the combination of both.

Table 2 Prediction Power and Accuracy of the Machine Learning Models Based on Different Input Parameters: GPS + IMA-Derived, FMP-Based Model or the Combination of Both

Input variable	R^2	MAE, U/L	RMSE, U/L	MAPE, %
GPS only	.619	85.605	141.148	0.332
FMP only	.601	88.242	144.563	0.346
GPS + FMP	.624	85.258	140.342	0.332

Abbreviations: FMP, Football Movement Profile; GPS, Global Positioning System; IMA, Inertial Movement Analysis; MAE, mean absolute error (in units per liter); MAPE, mean absolute percentage error (%); R^2, coefficient of determination; RMSE, root-mean-square error (in units per liter).

Feature Importance of the Whole-Sample Prediction Model

Figure 3 shows the top 20 features of the whole sample prediction models. The feature importance is characterized by the SHAP values in a descending order.

Individual CK-Prediction Models

One hundred and sixty-one players had at least 20 data point pairs (load on actual day, and CK on actual and following day). The total number of individual player observations (where both CK and external load data were available) was 6289. The average number of data points per player was 39 (15.5). The prediction power and accuracy of the individual models using all data points were relatively low ($R^2 = .369$, MAE = 104.93, RMSE = 162.9, MAPE = 0.953). Comparing the prediction power and accuracy of the prediction model using the whole sample with the individual prediction models for players with >20 data point pairs, the general model performed better in most cases (R^2 better 118 times out of 161 and MAPE better 112 times out of 161).

Discussion

The aim of the present study was to build general and individual-specific machine learning models using morning CK data and training or match load to predict the following days' CK value in national team soccer players. The sample size (444 players and 8570 individual observations) allowed us to compare the prediction power and accuracy of general and individual-specific prediction models. Even though the inclusion of both GPS + IMA-derived and FMP parameters resulted in the highest prediction power and accuracy, the negligible differences in the MAPE question whether building models using hundreds of parameters has real added value for practitioners. Therefore, monitoring 6 FMP parameters might be a good alternative in daily practice. Interestingly, in opposition to previous findings,[15,16] the general models outperformed the individual-specific CK prediction models in most cases.

Monitoring skeletal muscle damage in elite soccer can help practitioners in the weekly periodization and is more important in national team training camps, where congested schedules often occur. One of the most widely used biomarkers to monitor muscle damage and fatigue is CK,[6,29] which is sensitive to match-play

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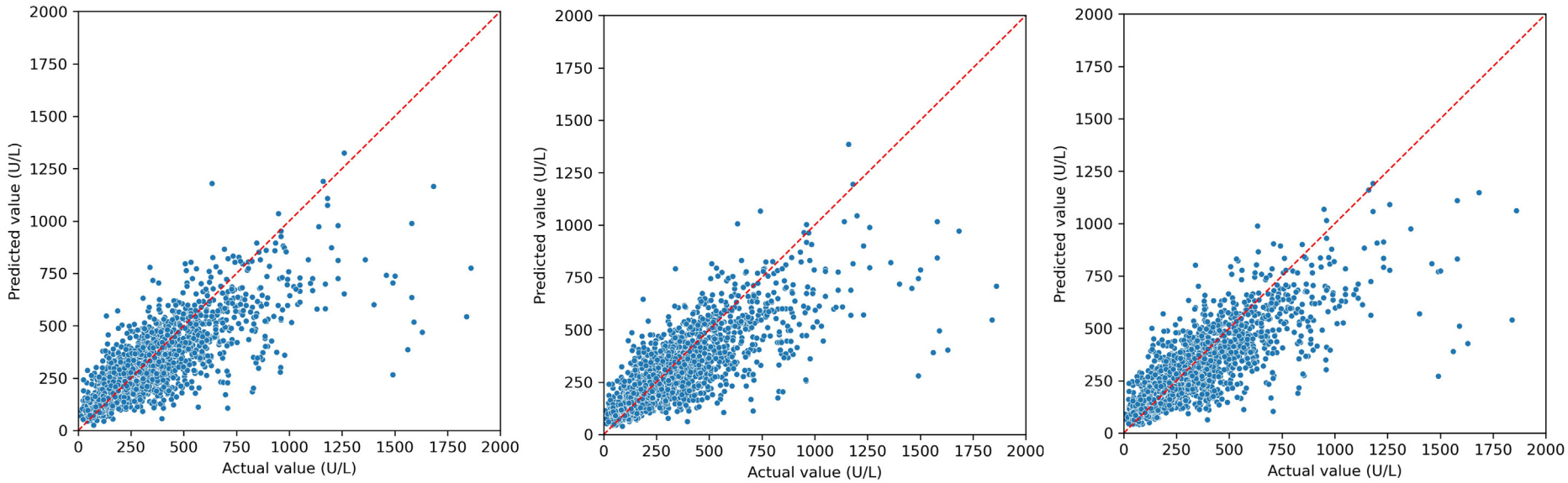

Figure 2 — Actual versus predicted creatine-kinase values (in units per liter) using (left) Global Positioning System– plus inertial-movement-analyis-derived input variables, (center) Football Movement Profile–based parameters, or (right) the combination of the 2. The figures provide a visual representation of the predictive power (R^2) of the machine learning models using 3 different input-parameter sets using our test sample (2571 data points).

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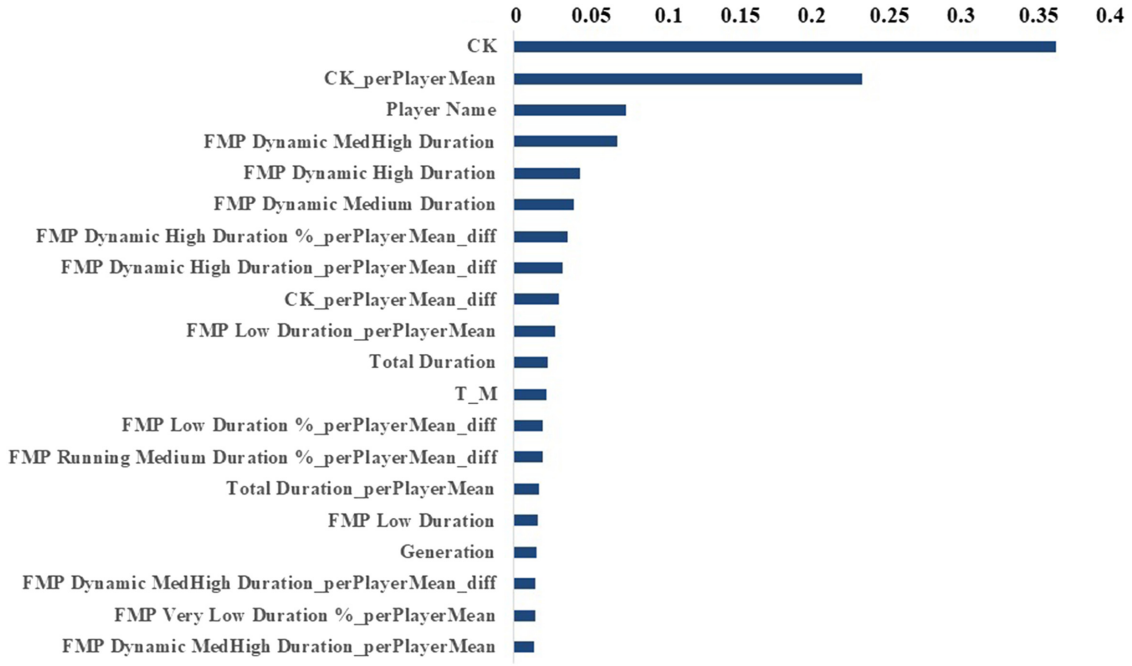

Figure 3 — The feature importance was characterized by the shapley additive explanations (SHAP) value, which quantifies the contribution of each feature to a particular prediction outcome. Parameters ending with "perPlayerMean" indicates individual average value from all historical data for a particular parameter; parameters ending with "perPlayerMean_diff," difference between the actual individual training/match parameter and the individual historical average for the same parameter; T_M, training or match; generation, the birth year of the player; CK, creatine-kinase; FMP, Football Movement Profile.

demands in elite soccer players.[30,31] Pregame CK levels have been shown to be increased in multimatch weeks in comparison to single-match weeks, if the player played at least 60 minutes in the first game,[32] highlighting its sensitivity to congested schedule. Elevated CK values compared to individual normative values have been suggested to result in 2.1% to 5.5% decreased training output in terms of high-intensity, very high-intensity, and explosive distance, maximal velocity, accelerations, and decelerations.[5]

Initial Model Using the Whole Sample

The initial results showed that the machine learning model best predicted the absolute CK value of the following day using the actual days' training/match load and CK value in the whole sample. Therefore, prediction of absolute and relative CK changes was not used in the refinement of the model. This is in agreement with a previous study, where a similar machine learning approach was used to predict the following days' CK value for different positions in elite youth soccer players.[15] On the other hand, absolute CK change showed the best correlation with FMP parameters in elite soccer players in another recent study.[17] However, that study used linear correlation and therefore cannot be compared with the current methodology.

GPS-Derived Versus FMP Parameters to Predict the Following Days' CK Value

GPS systems are commonly used in elite soccer to monitor external load during training sessions and matches.[7] In addition to GPS-derived metrics, most of these measurement tools also provide inertial sensor data (ie, accelerometer, gyroscope, and magnetometer). One of the advantages of the latter is that they do not need connections to the satellites; therefore, these parameters are available in closed stadiums and indoors as well. The FMP algorithm is based on inertial sensor data and assigns all soccer-specific movements into 6 categories. This is based on direction/movement type (running-linear and dynamic-change of direction) and intensity (very low, low, medium, and high) resulting in the following parameters: very low duration, low duration, running medium duration, running high duration, dynamic medium duration, and dynamic high duration.[17]

Using both GPS + IMA-derived and FMP or GPS + IMA only input variables provided better prediction power than the FMP parameters in isolation (R^2 = .62, .62, .60, respectively). However, the difference in prediction accuracy, as indicated by the MAE values using the 3 different input data sets (85.26 vs 85.61 vs 88.24 U/L) and the RMSE (140.34 vs 141.15 vs 144.56 U/L) was negligible. In one of the previous studies, position and activity type-specific (training, compensation training, and match) relationships were found between training/match load and the following days' CK.[15] In some cases, the prediction power of the machine learning models (eg, midfielders and wingers after training sessions; defenders and wingers after matches) was better than in the current study. However, in that study, all available GPS metrics (~1000) were utilized for the CK prediction. This fact limits the practical application of the previous results, as in some cases, the machine learning models included metrics in the prediction which are either practically irrelevant (eg, time spent >7 km/h, minimum effort duration 0.7–7 km/h, and distance 0–0.7 km/h) or not available through live monitoring (IMA parameters). This represents an advantage of the 6 FMP categories, as they are easily understandable for practitioners and can be monitored live. The latter allows immediate feedback to the coach, and adjustments can be made during training sessions or matches.

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Population-Based Versus Individual-Specific Prediction Models

One hundred and sixty-one out of 444 players had at least 20 data-point pairs (training/match load on a given day, and CK on that day and the following day). For them, individualized machine learning models were built. We assumed that these would be more accurate than the general model based on the whole sample. Interestingly, the average prediction power of the individual models was lower (R^2 = .369 vs .60–.62) and the MAE higher (104.93 U/L vs 85.61–88.24 U/L) than those of the general models based on FMP, GPS-derived parameters, or the combination of both. This is in contrast with previous findings, where individualized CK prediction models performed better than the general one in 25, U16-17 national team soccer players (R^2 = .64–.69).[16] Several methodological differences might explain the discrepant findings. The average number of data points per player was higher in the previous study than in the current one (57 [17] vs 39 [15.5]). Interestingly, there was no clear relationship between the number of data points and the prediction power of the models in the actual study. For players with >70 data points, the R^2 value was higher than below this threshold; however, they still did not reach the prediction power of the general models. Further studies are required to explain these findings. It should be noted that for 7 out of 25 players, the prediction power (R^2 was well below .5 in the previous study.[16] The current sample is larger (444 players and 8570 data points) than in the previous work (25 players and 1418 data points) and in most of the previous studies.[8–16] Therefore, we assume, that in smaller samples (~1500 data points in total, but with relatively higher number of data points per player [~60]), the individualized prediction models might work better.[16] On the other hand, if the number of data points is higher (>8000) without a concomitant increase of the data points per player (~40), the general model might be able to describe the relationship between the training/match load and the CK better, thus resulting in better prediction power. Finally, it should be noted that the model was given some additional input variables to increase its accuracy: load type (training/match), name of the player, position (goalkeeper, central defender, fullback, central midfielder, winger, and attacker), and age group. Using this input information, the general model performed slightly better (R^2 = .60–.62, RMSE = 140–144 U/L vs R^2 = .597, RMSE = 154 U/L) than the previous one with a much smaller sample size (8570 vs 1563 data points).[15]

Limitations

Our study has several limitations. First, the age range of the participants of the study was quite wide (range: 14–33 years). Previous results have shown that skeletal age influences CK recovery.[33] However, most of the U15 players in our sample were post peak-height velocity; therefore, biological maturity would be expected to have a limited impact on the results. In addition, our sample size (444 players and 8570 individual data points) could compensate for the effect of the age differences within the sample. Second, the CK measurement systems were changed during the study period due to external factors. To develop a correction equation between CK values provided by the 2 systems, 116 individual samples from 45 youth and adult soccer players were used.[23] In addition, the machine learning models had the best prediction power when merging the data sets provided by the 2 measurement systems. Third, even though a standardized food selection was offered in each training camp and CK was measured every morning after an overnight fast, we could not fully control individual nutrition which might have influenced the results. Fourth, a standardized recovery protocol was applied after the training sessions (recovery shake, 10-min ice bath in 12–15°C) and matches (recovery shake, 4 × 5-min contrast bath in 12–15°C and 36–38°C). As the recovery protocol was standardized, it could also have an effect on the CK change,[34] but this effect should have been comparable across age groups. Fifth, due to the applied setting, it was not possible to control the time between the last training session/match and the following days' CK measurement. Still, CK was determined each morning between 8 and 10 AM after overnight fasting. Finally, caution is needed when translating these results to a club schedule. During national team training camps, the program consists of 2 to 5 matches, with 1 to 3 days recovery in-between. Most of the CK data points were pretraining and posttraining sessions, where pretraining CK values were often higher than those of the following day. The tactical focus of the national teams' training sessions and the intention to maximize player readiness for the matches explains this trend. Further studies are needed to incorporate additional biomarkers (eg, myoglobin and lactate dehydrogenase), physical tests (eg, strength measurements, sprint test, and jumps), and or/subjective wellness and rate of perceived exertion into the machine learning models to provide a more complete picture of the recovery process after training sessions or matches.

Practical Applications

CK measurement can help to identify muscle damage, determine the recovery process after matches, and assist with periodizing the weekly microcycle in elite soccer players. Our results suggest that using machine learning models based on 6 FMP parameters results in similar prediction power and accuracy as using GPS + IMA-based prediction. The advantages of the FMP parameters are that they are easy to understand by practitioners, can be monitored live during training sessions and matches, and can also be used indoors and in large stadiums. Live monitoring enables immediate feedback to the coaches and, if necessary, quick adjustments to training. Machine learning models based on the whole sample (444 players and 8570 individual data points) predicted the following days' CK better than individualized models.

Conclusions

FMP, GPS + IMA-derived external load parameters from training sessions or matches, as well as the combination of both can be used to predict following days' CK in elite national team soccer players. General prediction models with a larger sample size than previously reported produced better prediction power and accuracy than individualized models. However, it should be noted that the general model had some additional inputs from the data points (load type [training/match], name of the player, position, and age group).

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